WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 2: Learning Health System

What is Life

- Erwin Schrödinger (1943) delivered influential lectures: “What Is Life? The Physical Aspect of the Living Cell and Mind” (published 1944)
- Persistence: functioning for ~3.8 billion years, despite catastrophes
- Explored key differences between living and non-living matter
- Proposed early hypotheses about the nature and molecular structure of genes ~10 years before Watson & Crick’s DNA model (1953)
- Highlighted that the “rules of life” appear to challenge simple physical intuitions about particle interactions
- Suggested living cells behave as if their molecules contain a form of “knowledge” or organization that sustains life
- **Stated that information (negative entropy) is the abstract concept that quantifies the notion of this order amongst the building blocks of life.**

What does Life Optimize?

Does it Optimize Development, Maintenance and Evolution of Systems?

Blueprint of Life

- Biology increasingly operates as an information science
- DNA **as information** in living systems
- The double helix structure was described in 1953 (Crick & Watson, 1953), received the 1962 Nobel Prize for discovering the molecular structure of nucleic acids and its role in information transfer
- 1953: DNA shown to be a double-stranded helix
- 1972: Recombinant DNA technologies enable cloning DNA
- 1977: DNA sequencing methods developed (Sanger & Gilbert)
- 1995: First complete bacterial genome: Haemophilus influenzae
- 2007: Complete human diploid genome sequences publicly announced
- 2009: First human genome sequenced using single sequencing technique
- 2010: Single-molecule (3rd gen) sequencing scales data from TB → PB range
- 2010: First publication of a human metagenome → new complexity

Big Picture of the Human Body

- A ~70 kg adult contains ~3×10^{27} atoms
- Built from ~60 chemical elements and ~10^{14} cells
- Cells = basic building blocks of life
- Contain supramolecular complexes, chromosomes, plasma membranes, etc.
- Built from macromolecules (DNA, proteins, cellulose) and monomers (nucleotides, amino acids)
- Core biology: gene → RNA → protein
- Central idea from DNA discovery to genome sequencing
- Yet the origins of health & disease remain poorly understood
- Interdependence: Microbiome of skin, mouth, eyes, and gut in healthy individuals differ remarkably in composition

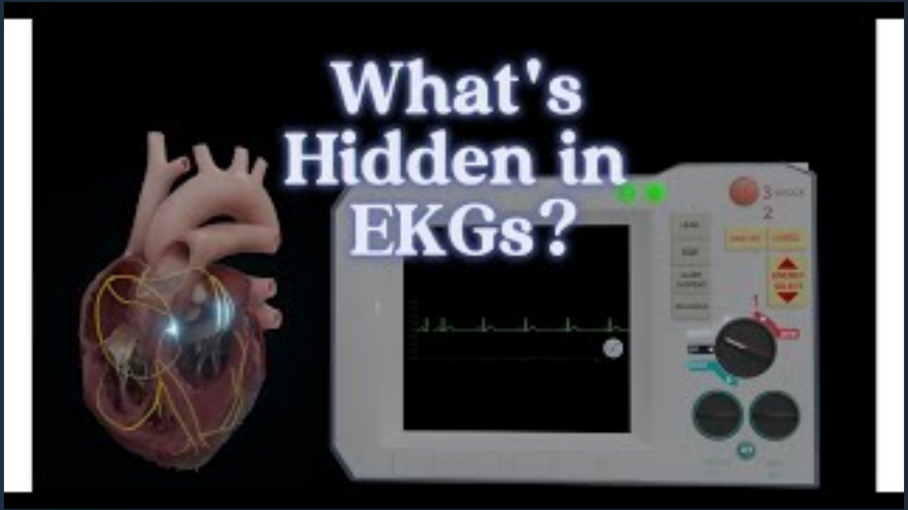
Tissues are Systems

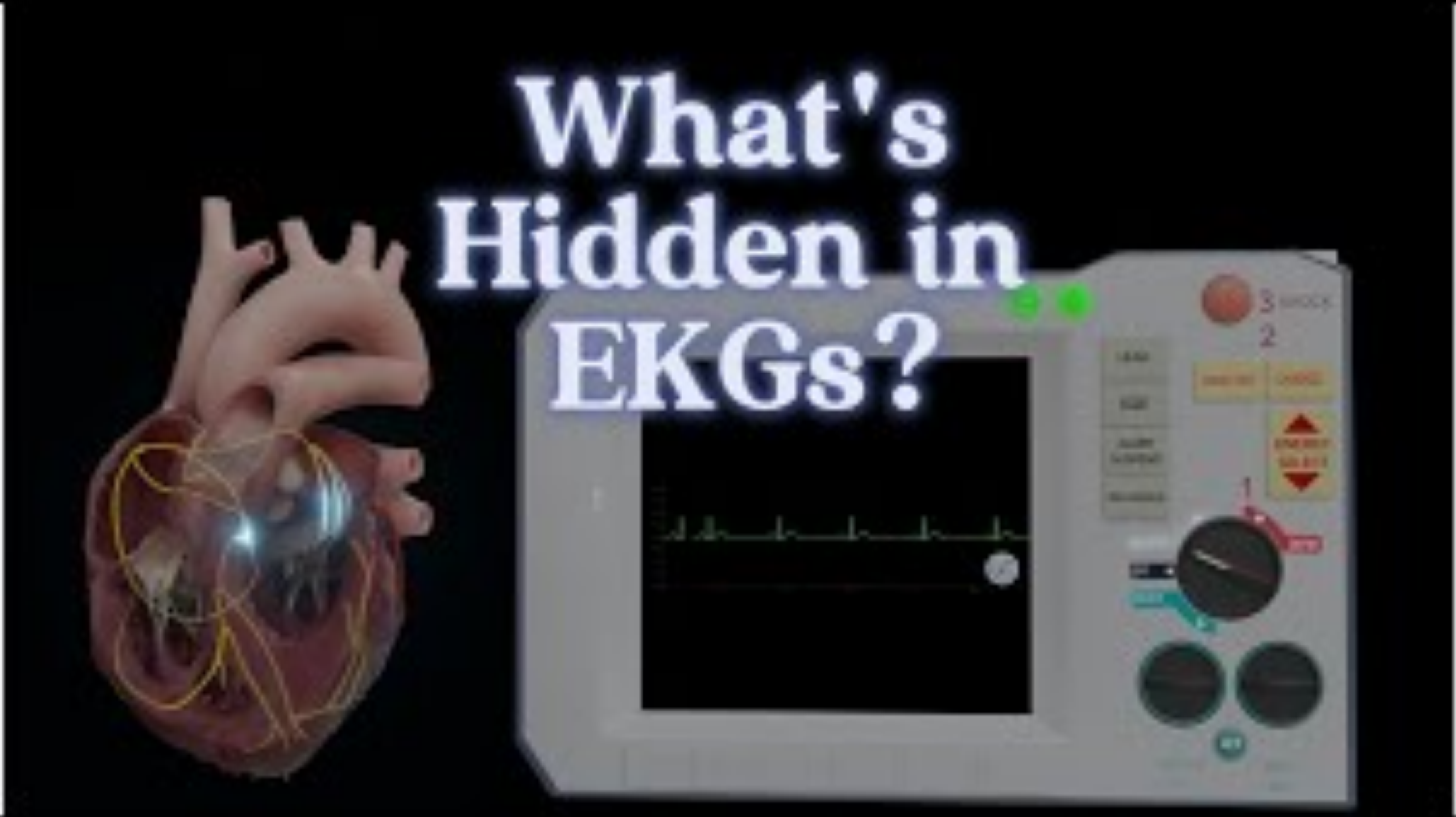
- A level of organization between cells and a complete organism
- An ensemble of similar cells (often from the same origin)
- Cells work together to perform a specific function
- Histology: study of tissues
- Histopathology: study of tissues in the context of disease
- Examples of major tissue types
  - Skin
  - Fibrous connective tissue (e.g., tendon)
  - Adipose tissue (fat)
  - Cartilage
  - Bone
  - Blood (white cells, red cells, plasma)

Organs are Systems

- The heart is a hollow muscle pumping blood throughout the blood vessels in the circulatory cardiovascular system, by repeated, rhythmic contractions.
- Organs are formed by the functional grouping of multiple tissues

https://www.youtube.com/watch?v=IiELdMPuR10

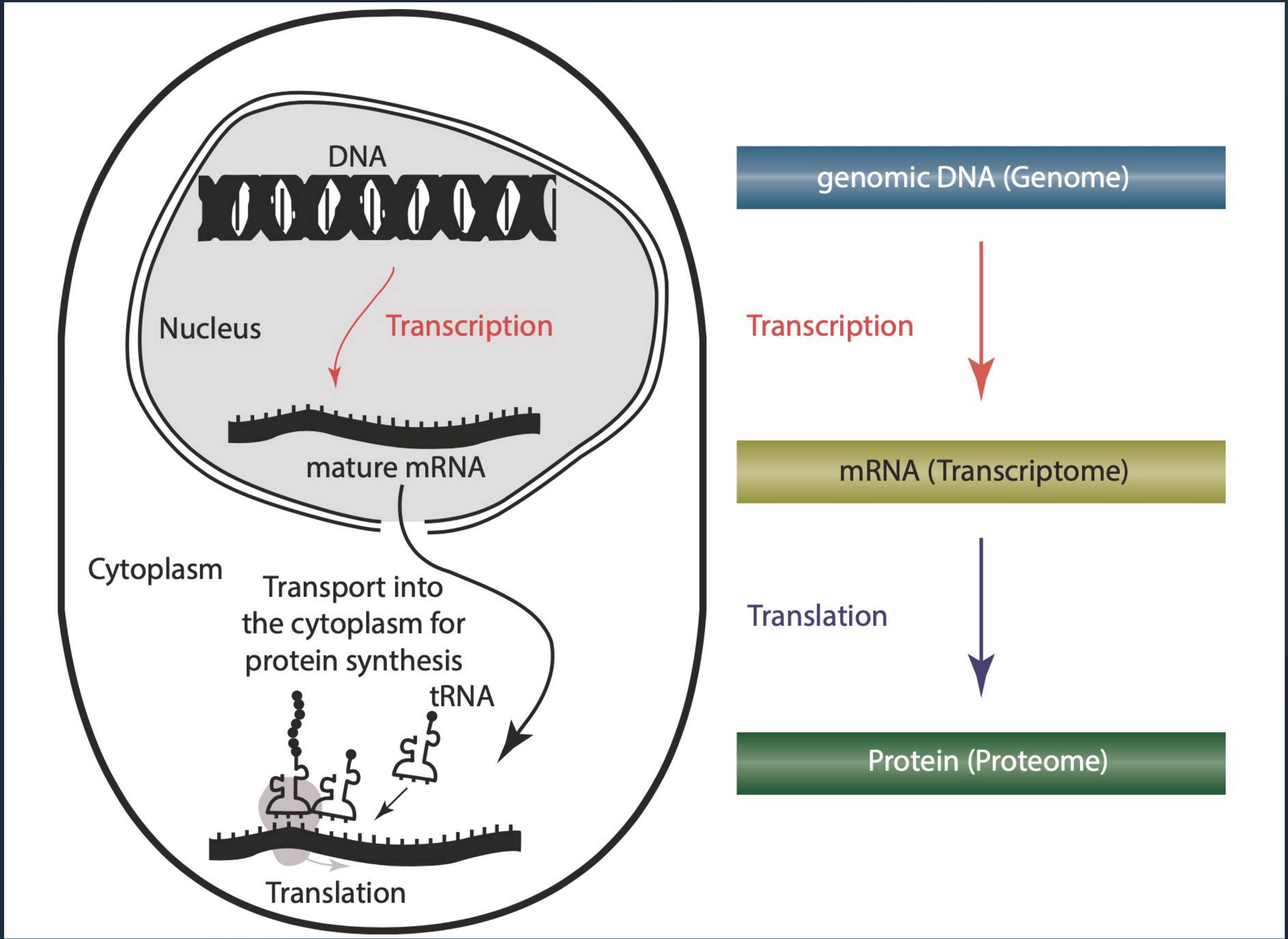



What's
Hidden in
EKGs?

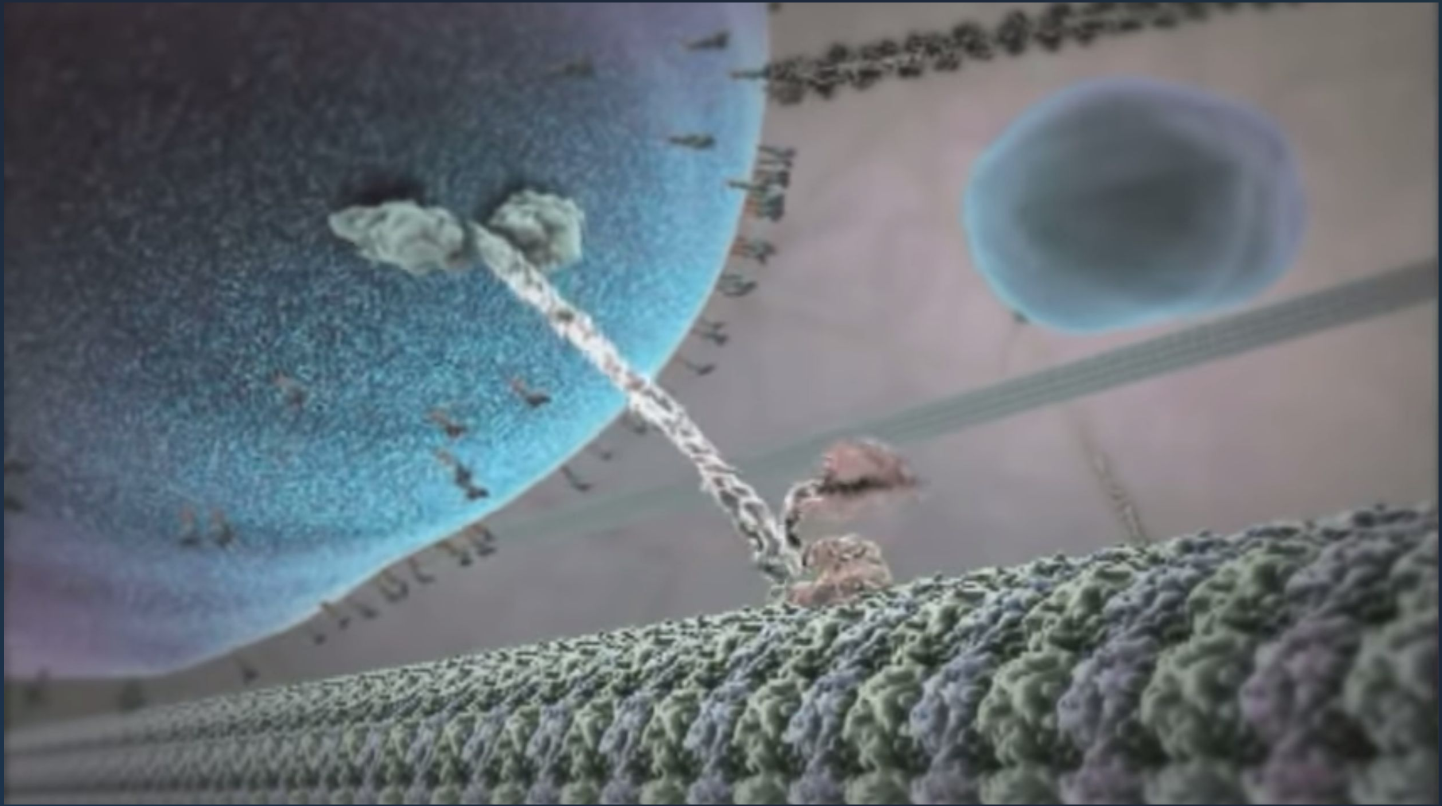
Human Body is a System of Systems

- Cardiovascular: heart + arteries + veins → blood circulation
- Respiratory: lungs + trachea → breathing and gas exchange
- Endocrine: hormones → communication and regulation
- Digestive: esophagus, stomach, intestines → digestion and nutrient absorption
- Urinary / Excretory: eliminates wastes and maintains fluid balance
- Immune / Lymphatic: defense against disease-causing agents
- Musculoskeletal: muscles + skeleton → stability, movement, support
- Nervous: collects, transfers, processes information (control + coordination)
- Reproductive: sex organs → reproduction

Central Dogma: A Micro System



Molecular Motors are "Systems"



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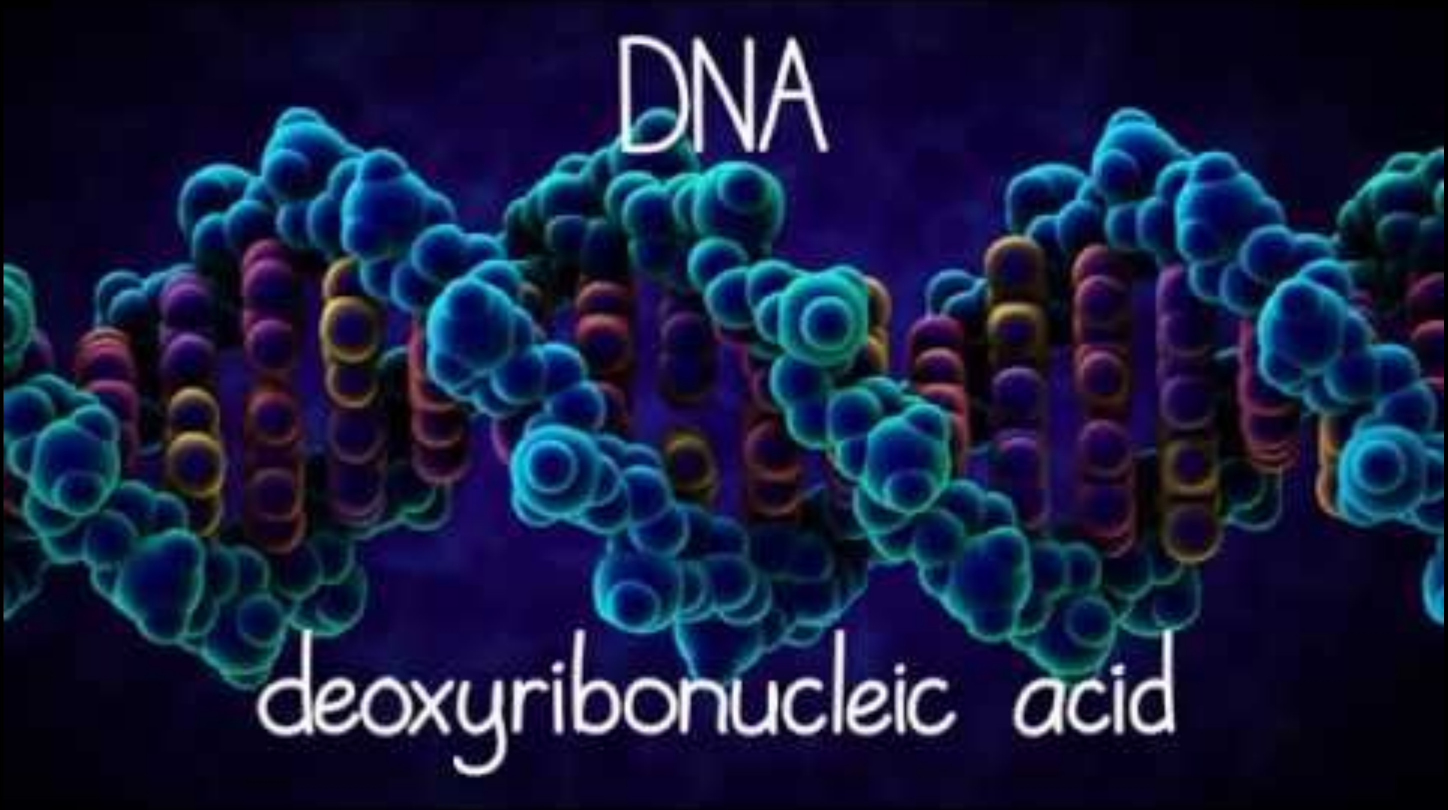
DNA is a System


DNA
deoxyribonucleic acid

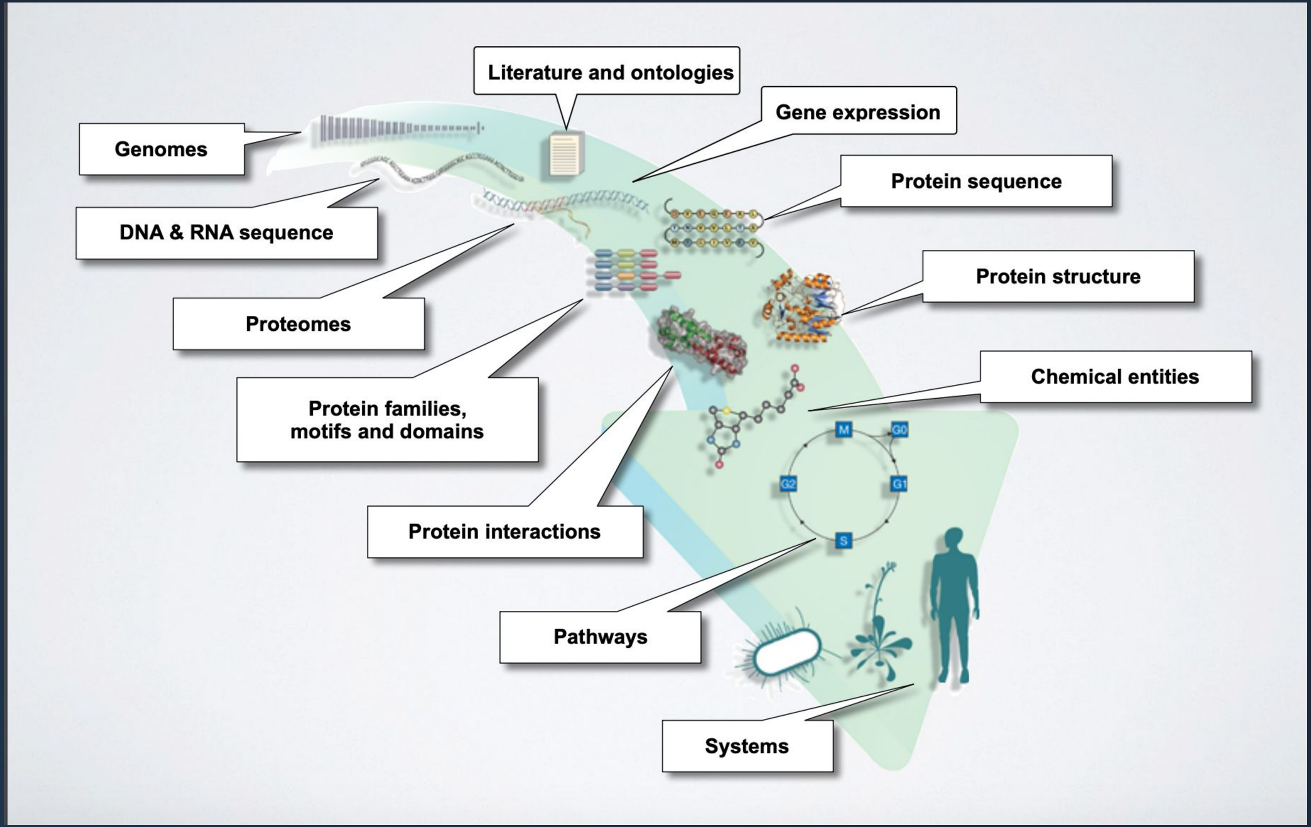
Rise of Complexity: “Omes” are Systems

Genome	The full complement of genetic information both coding and noncoding in an organism
Proteome	The complete set of proteins expressed by the genome in an organism
Transcriptome	The population of mRNA transcripts in the cell, weighted by their expression levels as transcripts copy number
Metabolome	The quantitative complement of all the small molecules present in a cell in a specific physiological state
Interactome	Product of interactions between all macromolecules in a cell
Phenome	Qualitative identification of the form and function derived from genes, but lacking a quantitative, integrative definition
Glycome	The population of carbohydrate molecules in the cell
Translatome	The population of mRNA transcripts in the cell, weighted by their expression levels as protein products
Regulome	Genome wide regulatory network of the cell
Operome	The characterization of proteins with unknown biological function
Synthetome	The population of the synthetic gene products
Hypothome	Interactome of hypothetical proteins

Definitions

Medical Informatics	1970 definition: "... scientific field that deals with the storage, retrieval, and optimal use of medical information, data, and knowledge for problem solving and decision making"
Bioinformatics	A discipline, as part of biomedical informatics, at the interface between *bio*logy and *infor*mation science and mathema*tics*; processing of biological data
Biomarker	A characteristic (e.g., body-temperature (fever) as a biomarker for an infection, or proteins measured in the urine) as an indicator for normal or pathogenic biological processes, or pharmacologic responses to a therapeutic intervention
Biomedical data	Compared with general data, it is characterized by large volumes, complex structures, high dimensionality, evolving biological concepts, and insufficient data modeling practices
Biomedical Informatics	2011-definition: similar to medical informatics but including the optimal use of biomedical data, e.g., from genomics, proteomics, metabolomics

Biomedical Informatics is the Study of These "Systems"



Informatics

- Informatics is the science of information, not simply the science of computers.
- Informatics is a young discipline — it has existed for only a little over 50 years.
- The term “informatics” comes from information + automatics (Steinbuch, 1957).
- It was coined to describe the science of automatic information processing.
- Informatics researches in and with computational methods, so it uses computers extensively.
- Informatics researchers don’t necessarily build computers or algorithms, although it is advisable.

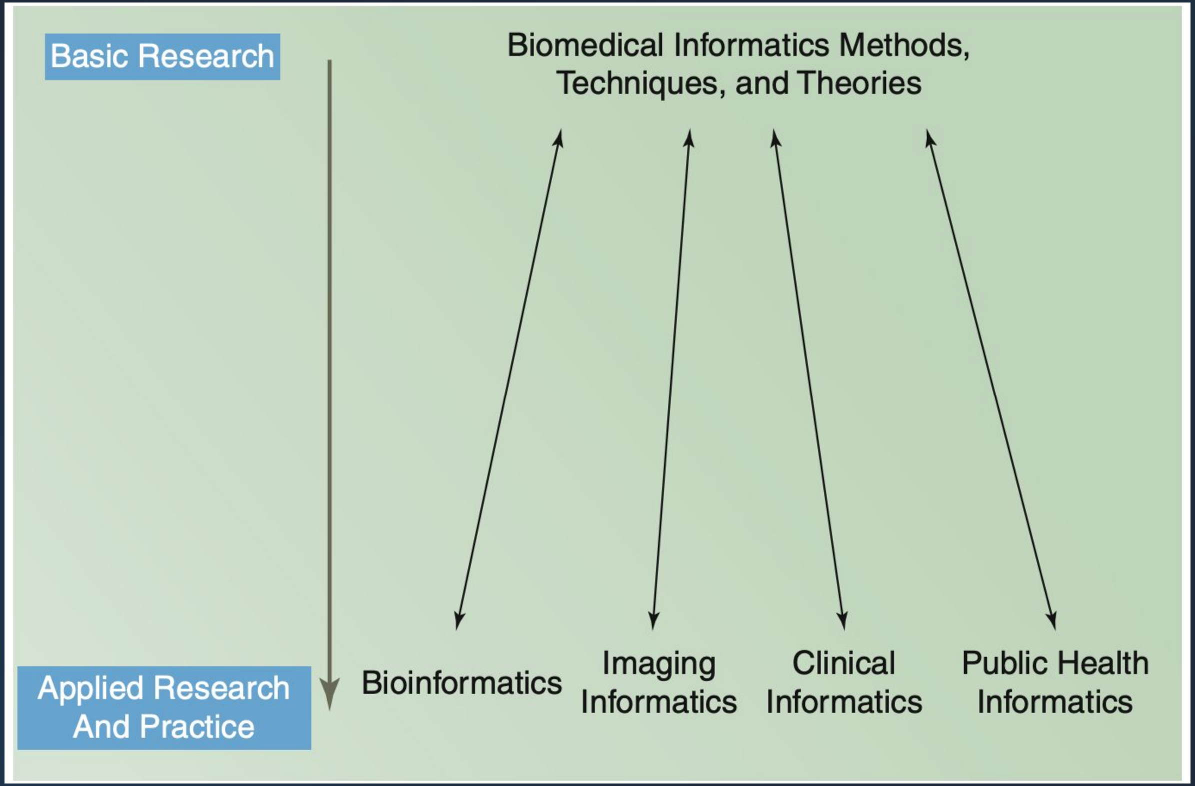
Biomedical Informatics

- Biomedical informatics is an interdisciplinary field that focuses on the effective use of biomedical data, information, and knowledge.

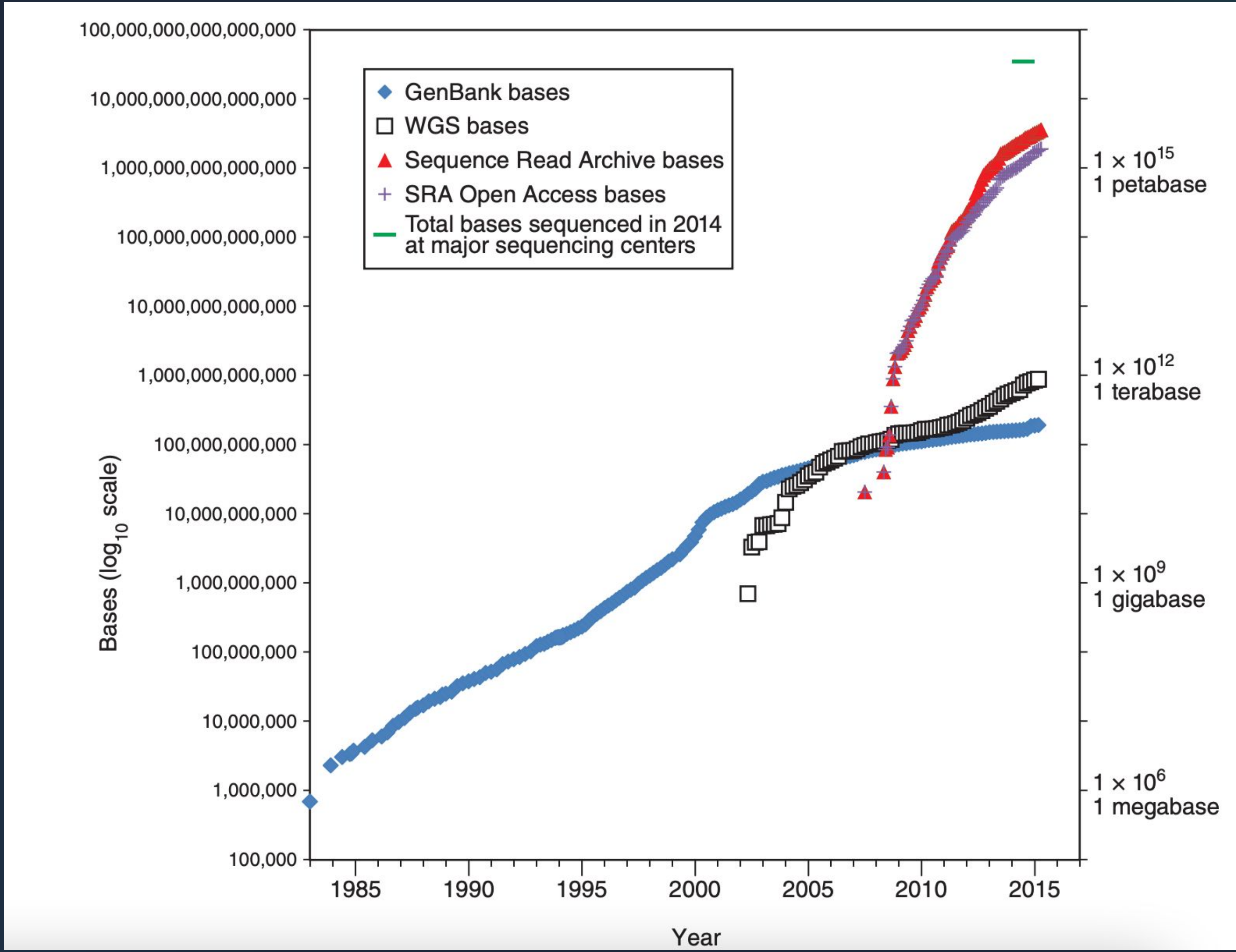
It supports key aims such as:

- scientific inquiry
- problem solving
- decision making
- The ultimate motivation is to improve human health (Shortliffe, 2011).
- Central idea: use machines to support and amplify human intelligence (Holzinger, 2013).

What is Biomedical Informatics



Eroom's Law


GenBank bases
WGS bases
Sequence Read Archive bases
SRA Open Access bases
Total bases sequenced in 2014 at major sequencing centers
Bases (log10 scale)
100,000,000,000,000,000
10,000,000,000,000,000
1,000,000,000,000,000
100,000,000,000,000
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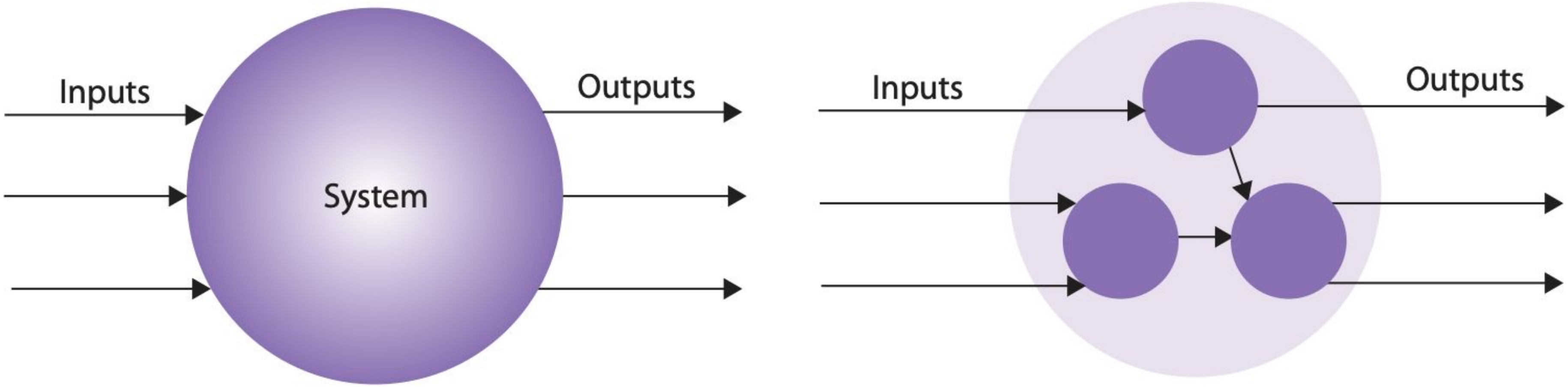
10,000,000
1,000,000
100,000
1 × 10^15
1 petabase
1 × 10^12
1 terabase
1 × 10^9
1 gigabase
1 × 10^6
1 megabase
1985
1990
1995
2000
2005
2010
2015
Year

Learning Health System



What is a Learning Health System?

- Routine care generates data → evidence → improved care (continuous feedback loop)
- Alignment needed
- science + informatics
- incentives
- culture
- governance/ethics
- Goal: best care at lower cost, continuously improving

Learnings

- Expect **adaptation + feedback loops**
- Monitor early and continuously (surveillance + rapid response)
- Use multi-pronged strategies (combination therapy, stewardship, infection control)
- Update models with real-world data (continuous learning)
- Build policies assuming **unknown unknowns** (resilience, not certainty)

Bottom line: *Act, but design interventions as experiments in a complex system.*

How Systems Work: Example from Antibiotic Resistance

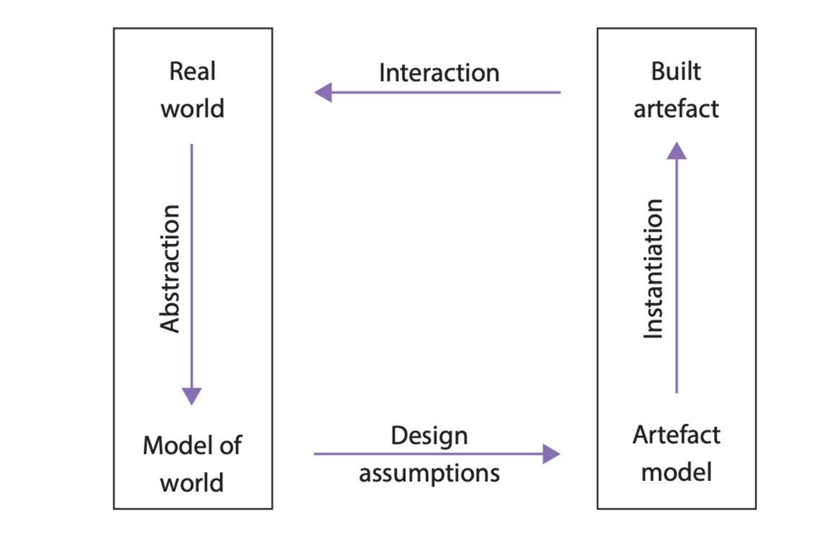
What happened

- Antibiotics created huge optimism: infectious disease looked “solvable.” Over time, bacteria evolved resistance (e.g., penicillinase producers).
- Antibiotics **selected for resistant survivors**, who then dominated populations.
- Some resistant infections now disrupt entire hospitals.

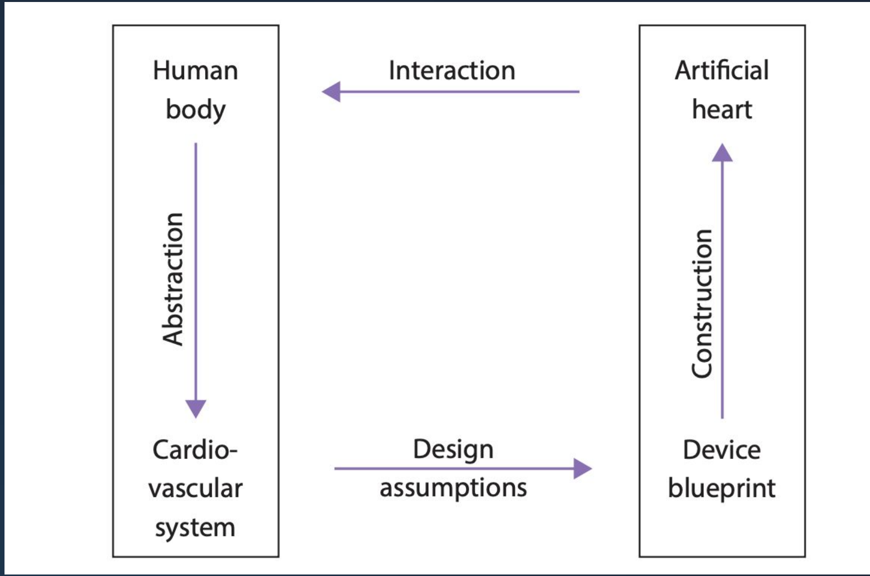
Why it happens (systems lens)

- Every intervention is based on a **simplified model** of reality. We act assuming “everything else is equal” → but systems never behave that cleanly. We rely on the **closed-world assumption**: “we know enough to act.”
- Resistance is not just a biology problem it’s a **predictable systems outcome** of intervention under **incomplete knowledge**.

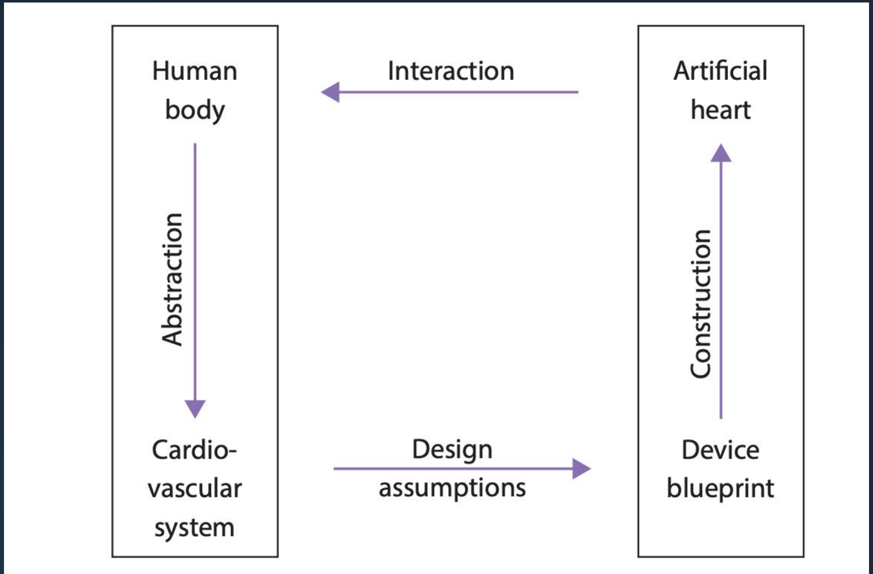
Models Reduce Complexity of Systems



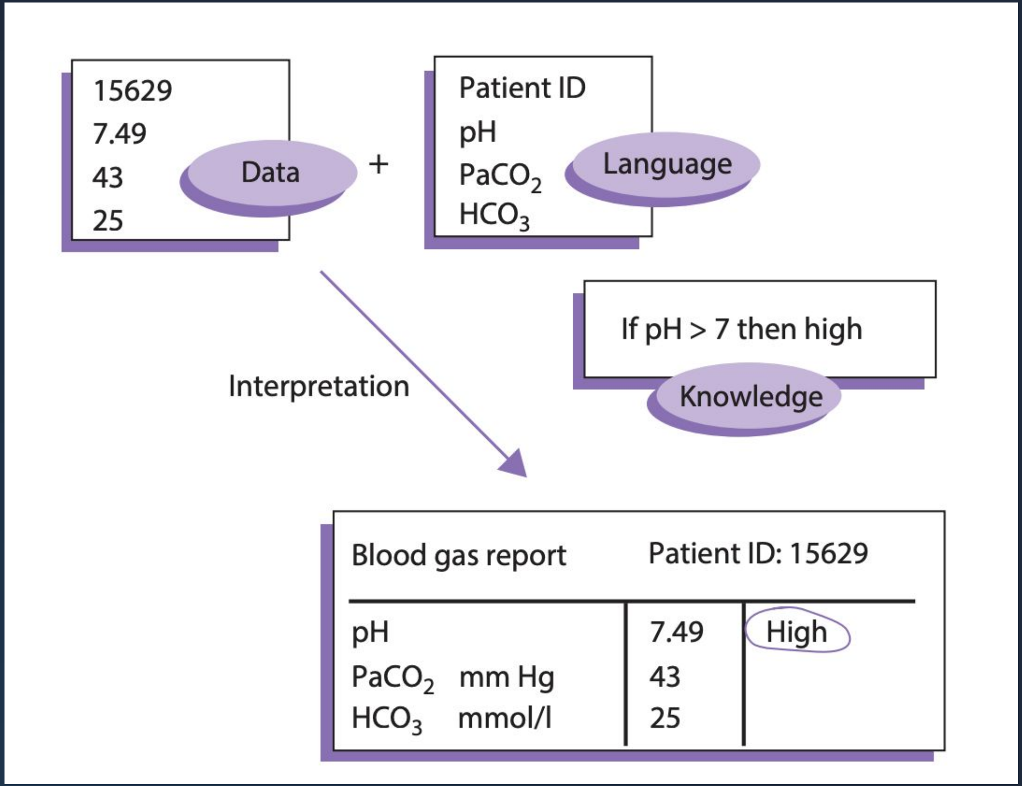
Act as Templates to be Instantiated to Simulate Systems



Models Create Real World Systems



Link Between Language, Data, Knowledge, Model



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